

Capstone Project: Plant Disease Detection

Role: AI Engineer

Core Technologies: Computer Vision, Multilabel-Classification, CycleGAN

Project Overview¹

Our team developed a plant pathology diagnosis application to detect and classify Korean crop diseases from a leaf image. The final production model successfully classifies 72 distinct classes, covering 50 different diseases and normal states across 22 types of agricultural crops.

Data Curation & Final Performance

Our team scaled the initially provided dataset (3 crops, 8 diseases) into a massive, robust dataset by sourcing additional data from AI HUB, Kaggle, and academic researchers.

- **Total Dataset:** Curated a final dataset consisting of 33,091 training images and 11,044 evaluation images.
- **Metric Selection (Macro-F1):** Due to the significant class imbalance across the 72 categories (e.g., 2,268 training samples for Red Bean Powdery Mildew vs. only 113 for Sesame Powdery Mildew), the **Macro F1-score** was chosen as the primary evaluation metric. By calculating the metric independently for each class and averaging them equally, this ensures the model's performance isn't skewed by majority classes and reliably detects even the rarest diseases.
- **Performance:** Achieved an overall macro average F1-score of 0.971, with near-perfect (1.00) F1-scores on critical crops like Apple, Eggplant, Strawberry, Tomato, and Watermelon.

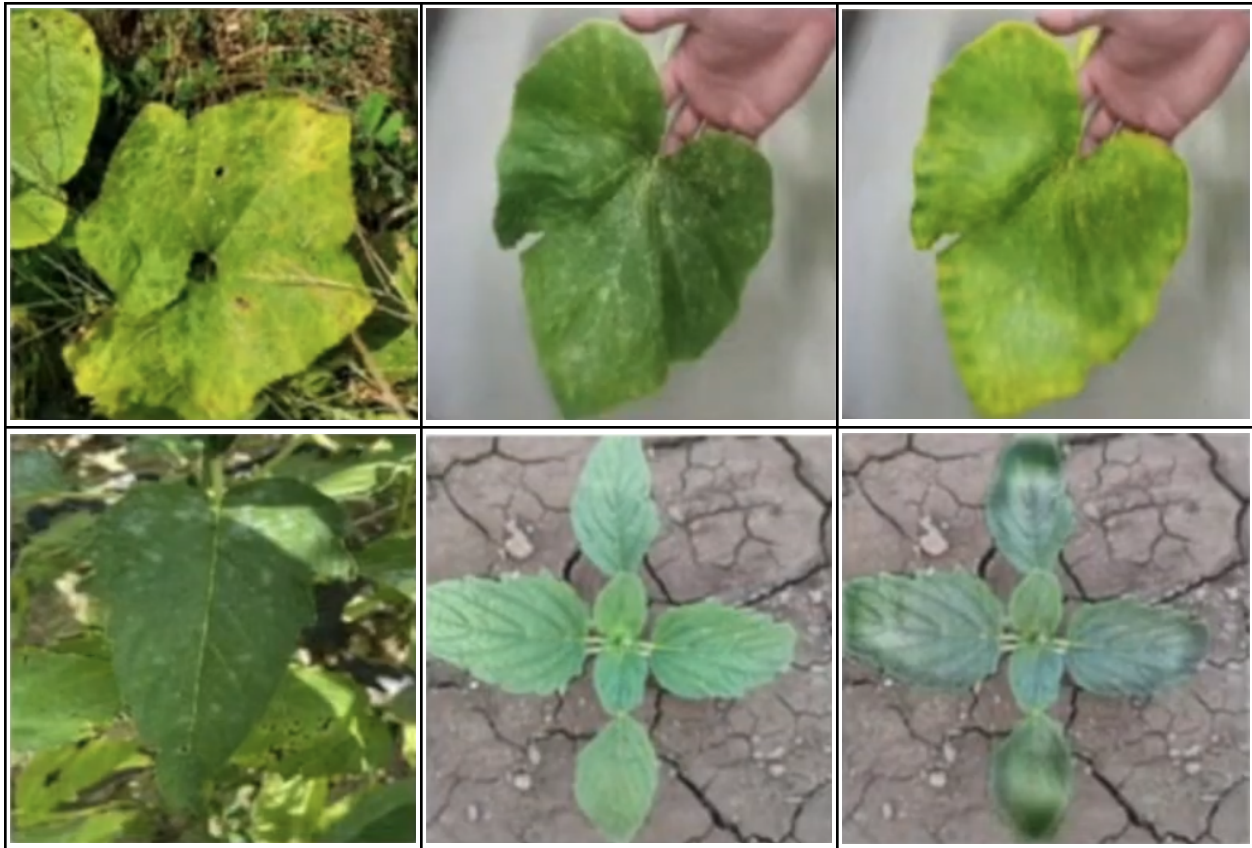
Key AI Methodologies & Contributions

CycleGAN² for Synthetic Data Generation

Disease Data	Normal Data	Generated Disease
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¹ <https://github.com/JeongjunBae/debug-AI/blob/main/README-EN.md>

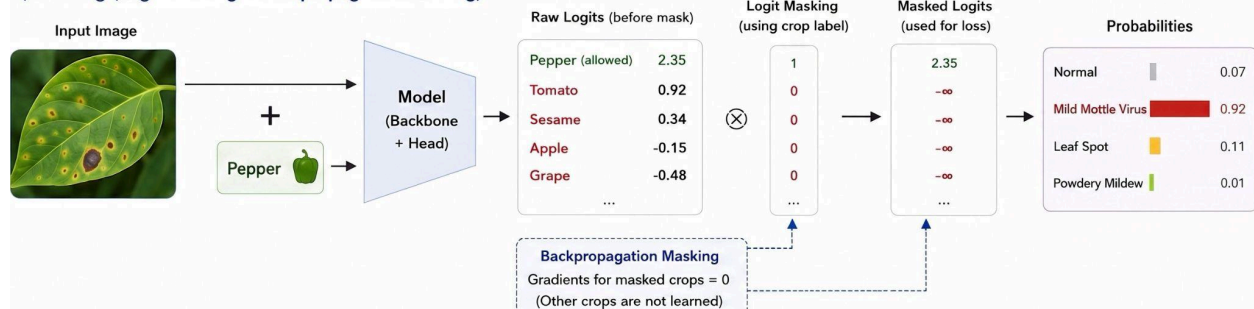
² <https://arxiv.org/abs/1703.10593>



Addressed the high cost and logistical difficulty of acquiring diseased plant images (which typically require isolated environments and expert virus injection). CycleGAN generates realistic diseased images from cheaper, more accessible normal images

Crop-aware Architecture

1) Training (Logit Masking + Backpropagation Masking)



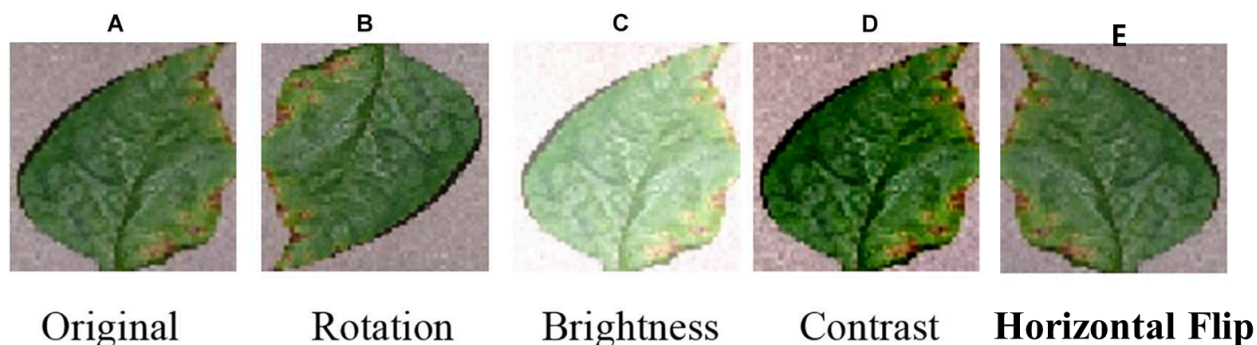
2) Inference (Logit Masking)



Capitalized on the premise that identifying the base crop type is trivial compared to diagnosing the specific disease. Engineered a contextual filtering mechanism that feeds the crop type alongside the image during both training and inference.

- **Training:** During training, backpropagation for non-target crop types is zeroed out.
- **Inference:** During inference, the model masks the confidence scores of irrelevant crop types (setting them to 0), drastically reducing inter-class confusion.

Robust Data Augmentation



Designed a comprehensive augmentation pipeline to further improve model generalization on field data.

Ablation Study & Optimization³

Conducted a rigorous ablation study to validate the architectural choices. The baseline model achieved a 0.8729 Validation Macro F1-score. Integrating PlantGAN improved performance to 0.8971, while adding the Crop-aware architecture drove the score up to 0.9655. The final integration of the full augmentation pipeline resulted in the state-of-the-art 0.9710 Validation Macro F1-score.

Ablation Study Table

GAN	Crop-aware	Augmentation	Validation Macro F1-Score
X	X	X	0.8729
O	X	X	0.8971
O	O	X	0.9655
O	O	O	0.9710

Awards

Third Prize⁴: 2022 Sejong University Creative Comprehensive Design Competition.

Popularity Prize⁵: 2022 Sejong University Creative Comprehensive Design Competition.

³ https://drive.google.com/file/d/1snUUTO9Awm21kvphDfDeMOblA1Oxu7kQ/view?usp=drive_link

⁴ https://drive.google.com/file/d/1bahM5ottudMM88nuXQMMx_oCmH7-B7SA/view?usp=drive_link

⁵ https://drive.google.com/file/d/1aOuhs1cmdkzZmD5gbf9-kHjqTDATsRp/view?usp=drive_link